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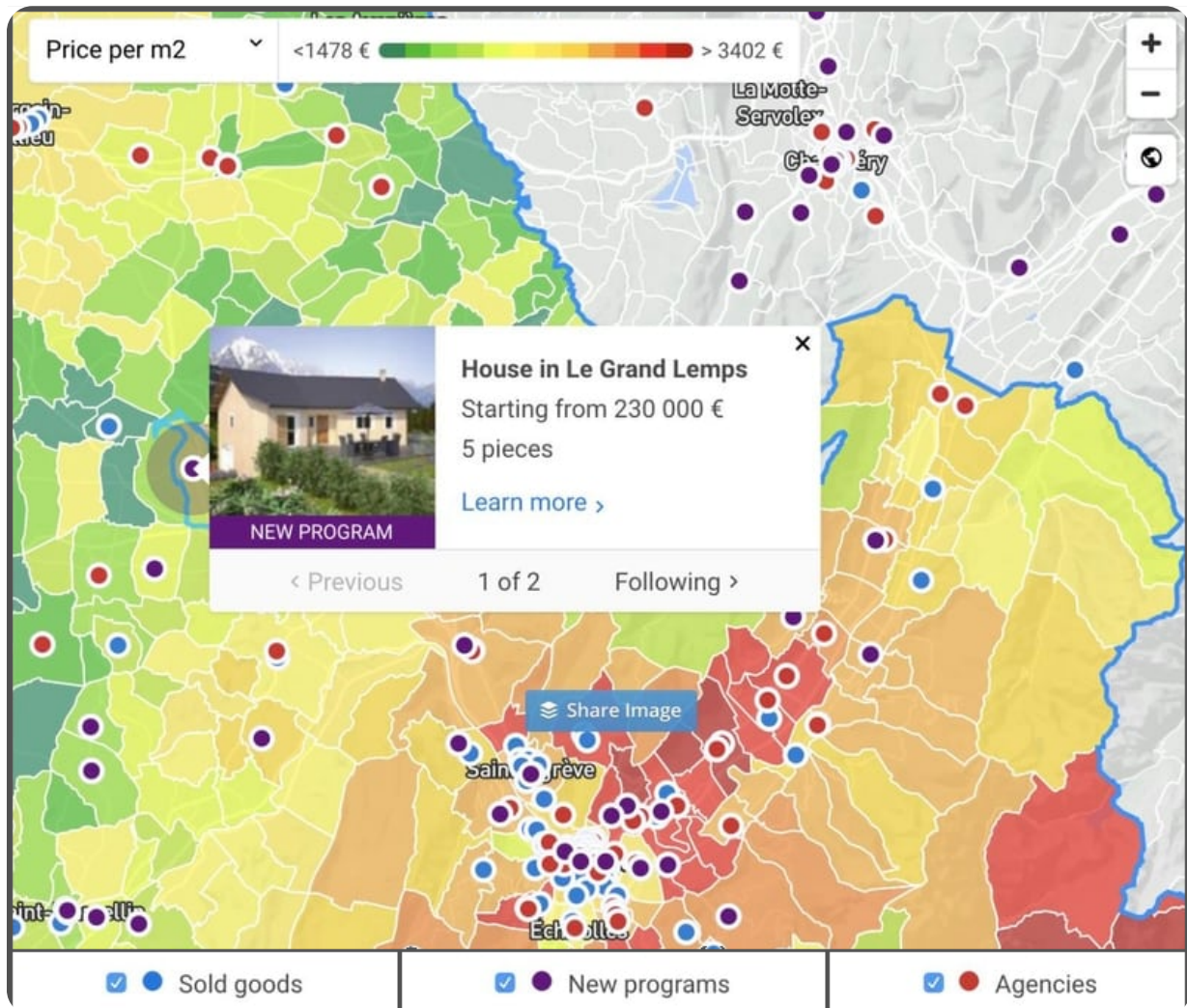
HOMEPEDIA

< EXPLORING HOUSING INSIGHTS />



HOMEPIEDIA

Your goal for this project is to develop an innovative app that provides users with comprehensive information and analysis on the housing market in France.



Through data collection, processing, and analysis, you will create a platform which offers a user-friendly and interactive experience, empowering users to explore and sift through the intricate dynamics of the French housing landscape.

Whether it's examining price trends, exploring regional variations, or delving into demographic factors, your app will serve as a tool for users to access the statistics and insights they desire...

...and even more: your app will provide so much value and functionality that it will significantly change the industry!

Specifications

You are expected to deliver a fully interactive app with **tabular** data, **cartographic** data, and data inferred from **textual** analysis.

The app integrates with your database(s) and leverages the **cluster of machines** employed for data analysis.

To provide a comprehensive understanding of your project's foundation, a well-defined schema of your database(s), along with a meticulous description of your data cleaning methodology, must be included.

In addition, if AI algorithms are utilized within the project, it is essential to provide a detailed account of their implementation and functionality.



Try using tools such as [streamlit](#) to create your app.

Goal

In this project, you will be doing **DataOps**.

Your main users won't be people looking for housing. They will be real estate firms, local governments, data analysts from big companies... Your app will help them do their job, which is to find answers to big questions like *"How can I reduce inequality in my department"* *"What are the risks of investing in this lot caused by global warming at a 15 years horizon"* *"Where should the next 10 franchises of my company be located?"*...

This doesn't mean that you shouldn't work on user experience. Your whole job is to make data easy to capture, parse, analyse... Reduce friction, and serve quality visualizations from which insights will be extracted.

Data

For this project, the following steps are mandatory:

- ✓ Data gathering
- ✓ Database organization
- ✓ Big data analysis
- ✓ Interactive visualizations

Data Gathering

Collect **extensive** data on French housing at various levels (state, region, department, city). Supplement this data with various metrics and indicators (economy, education, energy, environment, infrastructure, population, ...).



Some inspirational websites (Zillow, MeilleursAgents, ...) may restrict automated scraping in their terms of service. As this is an academic, non-commercial project, this is tolerated for the purpose of the exercise - but you should stay mindful of sourcing methods and favor open/public datasets (INSEE, data.gouv.fr, ...) where possible, and document your sourcing choices.

You should aim for at least a dozen data sources, and a maximum of 50.

At least one of these sources should be a *textual* one. This type of data mustn't be simply mapped to its source or presented as a list of comments, but it should be processed in a way that makes it usable.



How many times are *rats* mentioned in reviews in the local restaurants in this region?

While the primary focus lies in collecting and analyzing vast amounts of data, there is a crucial role for artificial intelligence to play. Textual data, such as comments describing the quality or shortcomings of a city, can be efficiently processed and analyzed using AI techniques, to extract valuable information such as sentiment analysis.

Database organization

Structure the data into suitable databases, employing **efficient indexing and optimization techniques**.

Your data **MUST** be organized in specific databases.

You have to create relational **and** non-relational databases in order to proficiently classify your tabular and non-tabular data.

Since it comes from different sources, **standardization** will be necessary.

Big data analysis

Make use of **advanced tools and methods** in order to extract valuable insights from the huge amount of data.

To tackle the immense volume of data involved in this project, you HAVE TO use technologies for **fast and efficient data processing**, such as *Hadoop* and *Spark*.

By creating a **cluster of machines**, you will distribute the workload across multiple nodes, significantly reducing processing time and enabling parallel data processing.



Look at cloud solutions, such as [Databricks](#): an independent platform built on Apache Spark, available on AWS, Azure and Google Cloud. Its [free edition](#) is enough to get started.



Talk to your pedago about which solutions can be found to make those technologies accessible to you. Some campuses will be given tokens, others might make use of local VMs.

Interactive visualizations

Present the results through an **interactive** app, using visualizations to provide an **intuitive** and **comprehensive** understanding of housing trends.

These steps are fundamental: your app effectively **investigates** and **represents** the data. Your outputs encompass traditional tabular presentations, showcasing statistical insights. Additionally, your app also incorporates cartographic analysis, allowing users to visualize data on [bubble | isoline | choropleth | heat | route | connection | ...] maps.

You have the freedom to design the app according to your preferences, but it MUST be **highly interactive and customizable**: your final users are able to navigate and select what they want to explore (areas, analysis results, global indicators, ...).

Another constraint about scale: the analysis MUST span various levels, including city, department and region.



Websites like [zillow](#), [meilleursagents](#), or [mapbox.com](#) can provide you some inspiration. Do not copy them, but take a look at them.

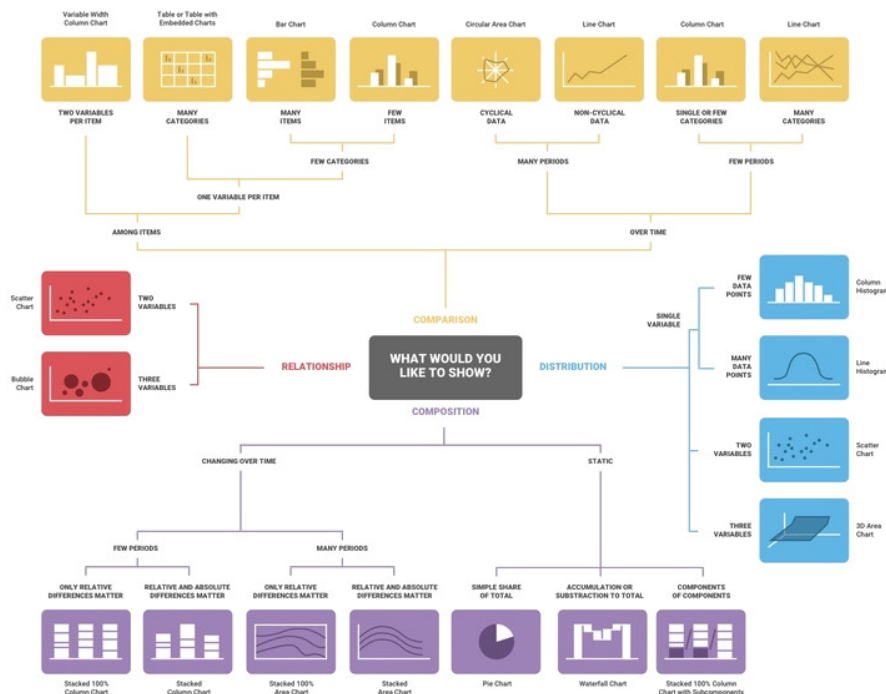
Tabular Data

Deliver sortable and filterable tables presenting the key statistical indicators of your dataset (price per m², price evolution, demographic and infrastructure metrics, ...) at the various levels of granularity (city, department, region). Users must be able to cross-reference indicators and, ideally, export the displayed data.

Cartographic Data



One of the trickiest parts of the analysis process is choosing the right way to represent your data, and to avoid any [chartjunk](#). Websites like [anychart](#), [datavizcatalogue](#), [from data to viz](#) or [datavizproject](#) may help you.



Tools like [leaflet](#) allow you to build interactive maps and integrate them in your app.

Bonuses

You can improve this project in many ways, including:

- ✓ including other countries housing markets;
- ✓ deploying your app online;
- ✓ adding an authentication protocol and accounts;
- ✓ increasing your application's security;
- ✓ building an admin view for the server;
- ✓ implementing real time updates;
- ✓ adding a guided tour;
- ✓ ...

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